

Cloud Computing Applications and Services

(Aplicações e Serviços de Computação em Nuvem)

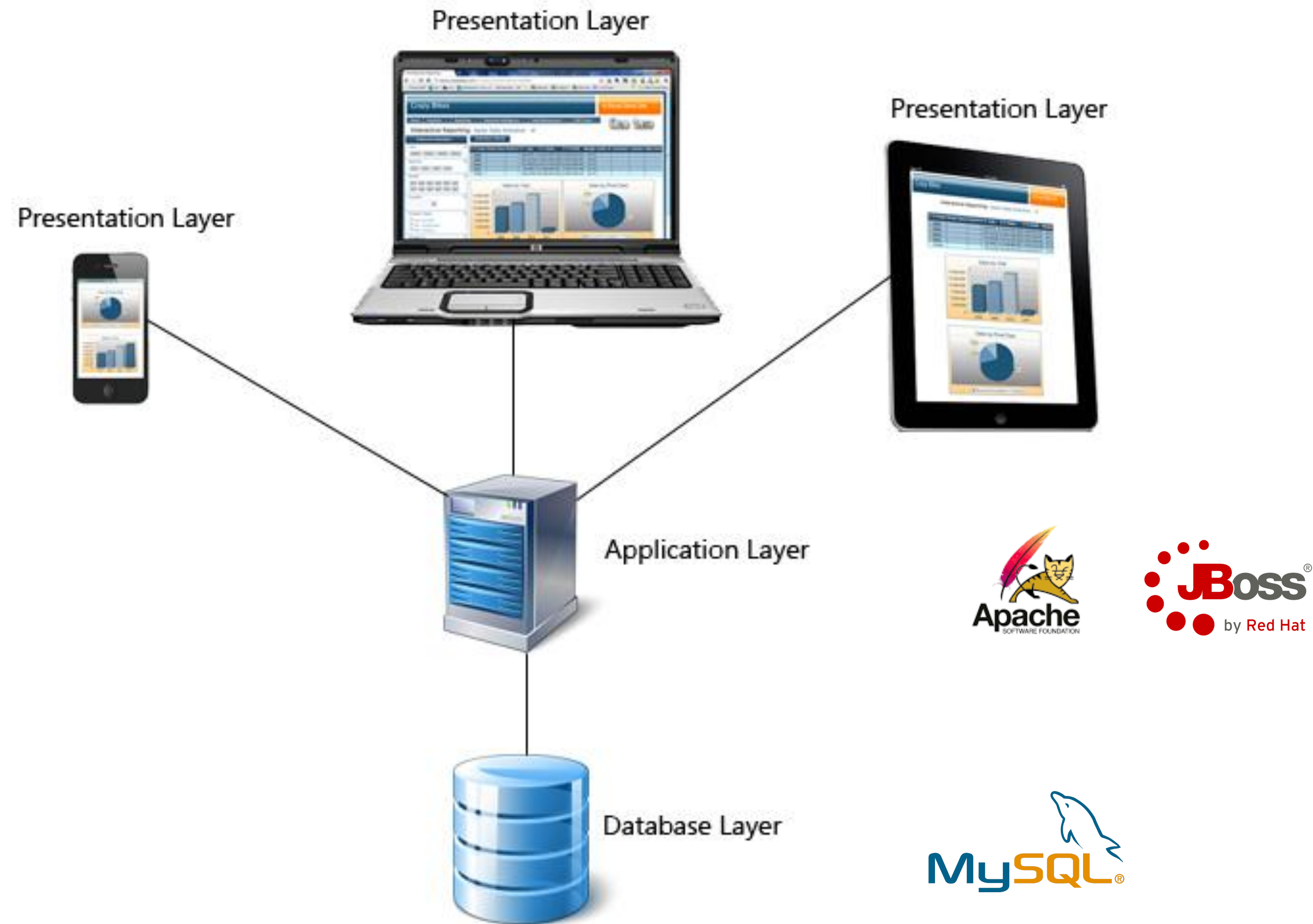
Introduction

University of Minho

2025-2026

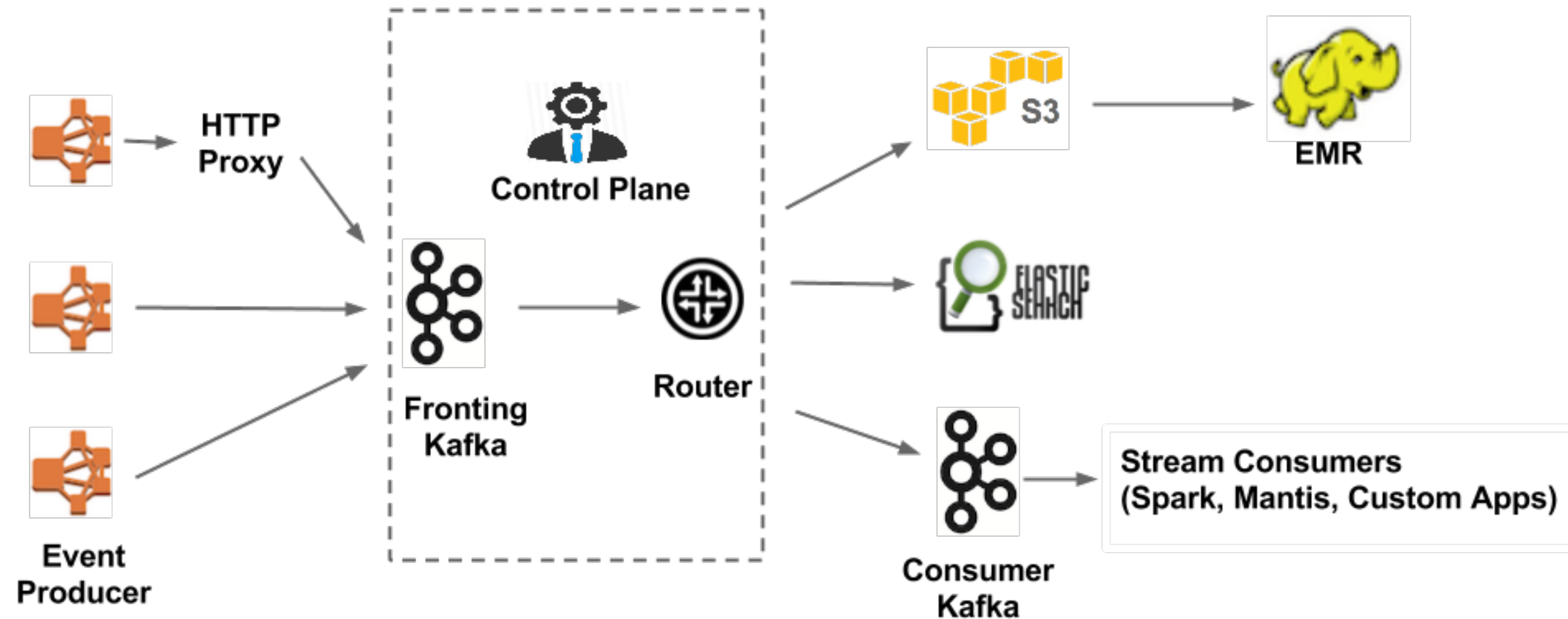


A “Simple” Application



Complex Applications

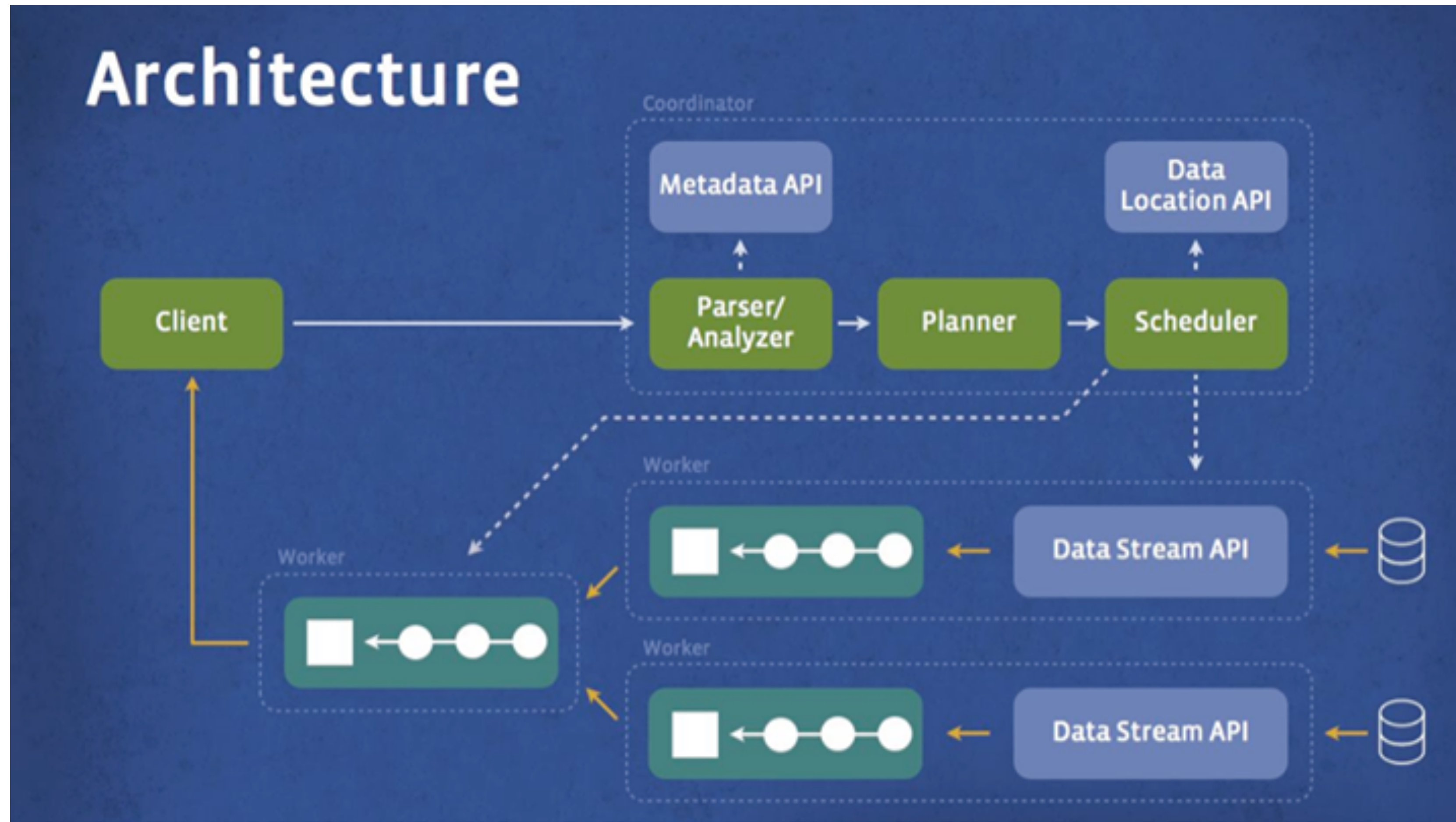
Netflix – Keystone Data Pipeline



Source: <https://medium.com/netflix-techblog/evolution-of-the-netflix-data-pipeline-da246ca36905>

Complex Applications

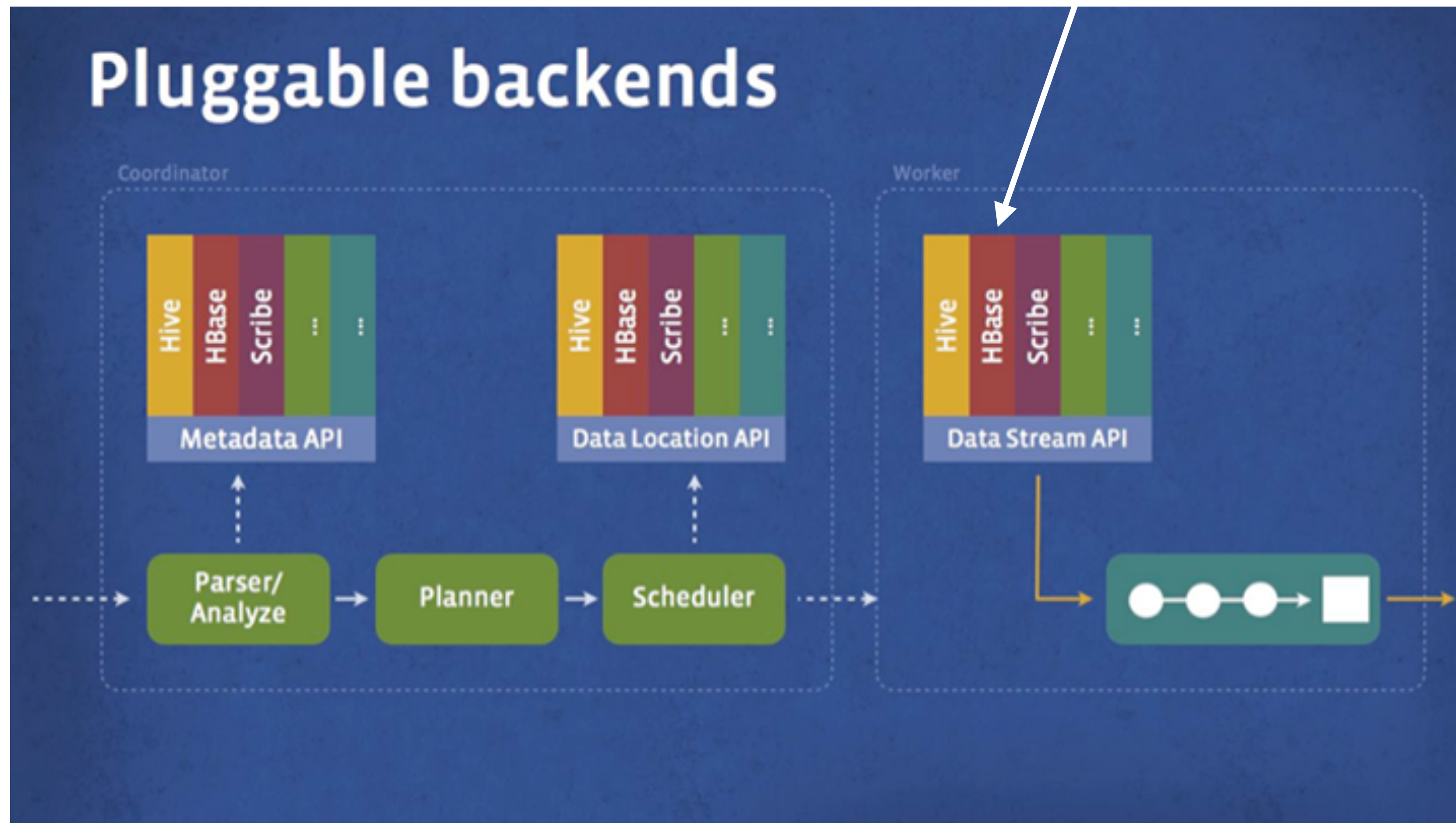
Facebook – Presto SQL Engine



Source: <https://www.facebook.com/notes/facebook-engineering/presto-interacting-with-petabytes-of-data-at-facebook/10151786197628920/?s=keen-io>

Complex Applications

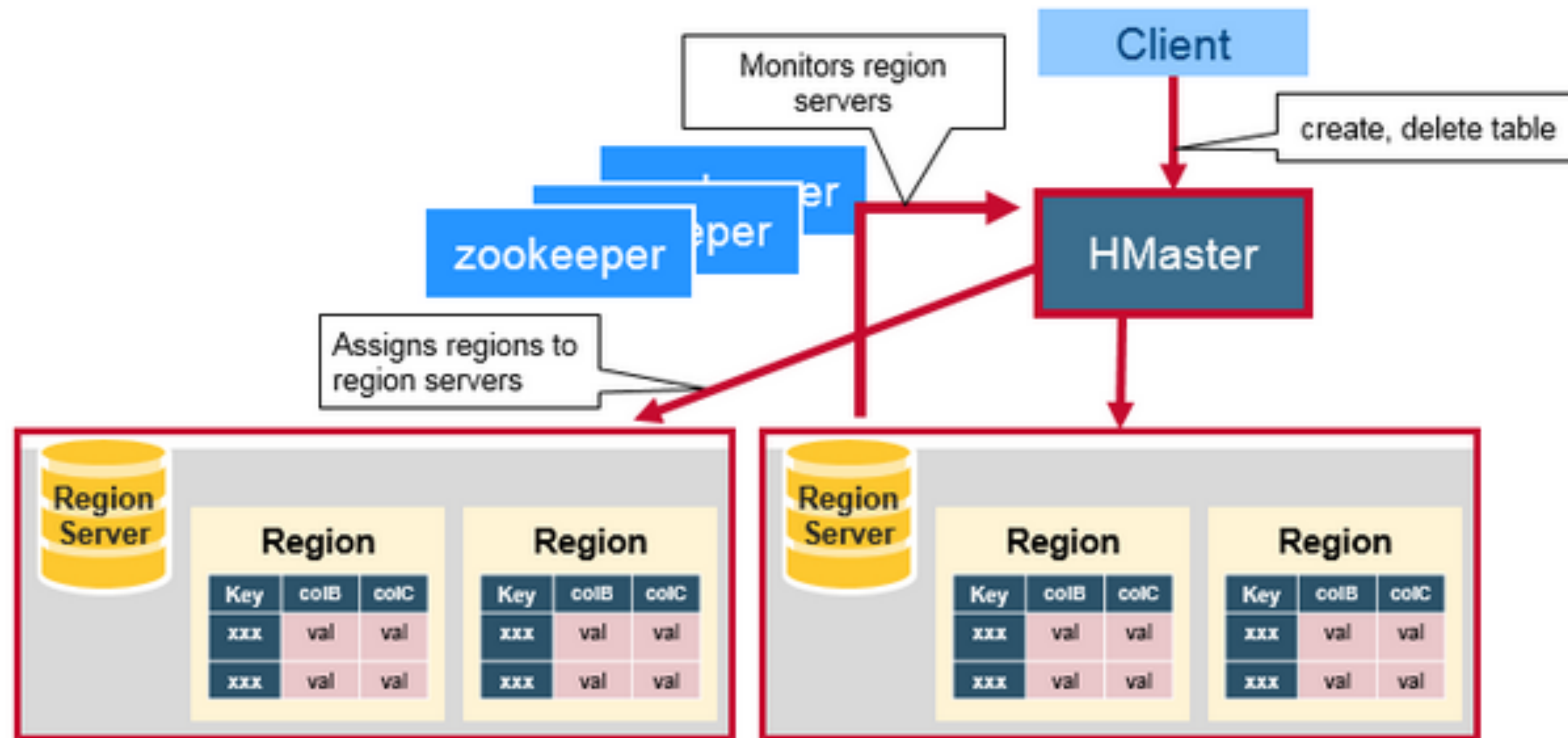
Facebook – Presto Backends



Source: <https://www.facebook.com/notes/facebook-engineering/presto-interacting-with-petabytes-of-data-at-facebook/10151786197628920/?s=keen-io>

Complex Applications

HBase



Complex Infrastructures



Challenges

● Deployment

- Applications have multiple components with different requirements (e.g., hardware, libraries)
- Heterogeneous hardware, software and services (e.g., HPC services, cloud computing services)



● Management & Configuration

- Specific to each software/hardware component
- Optimal configurations will change over time...

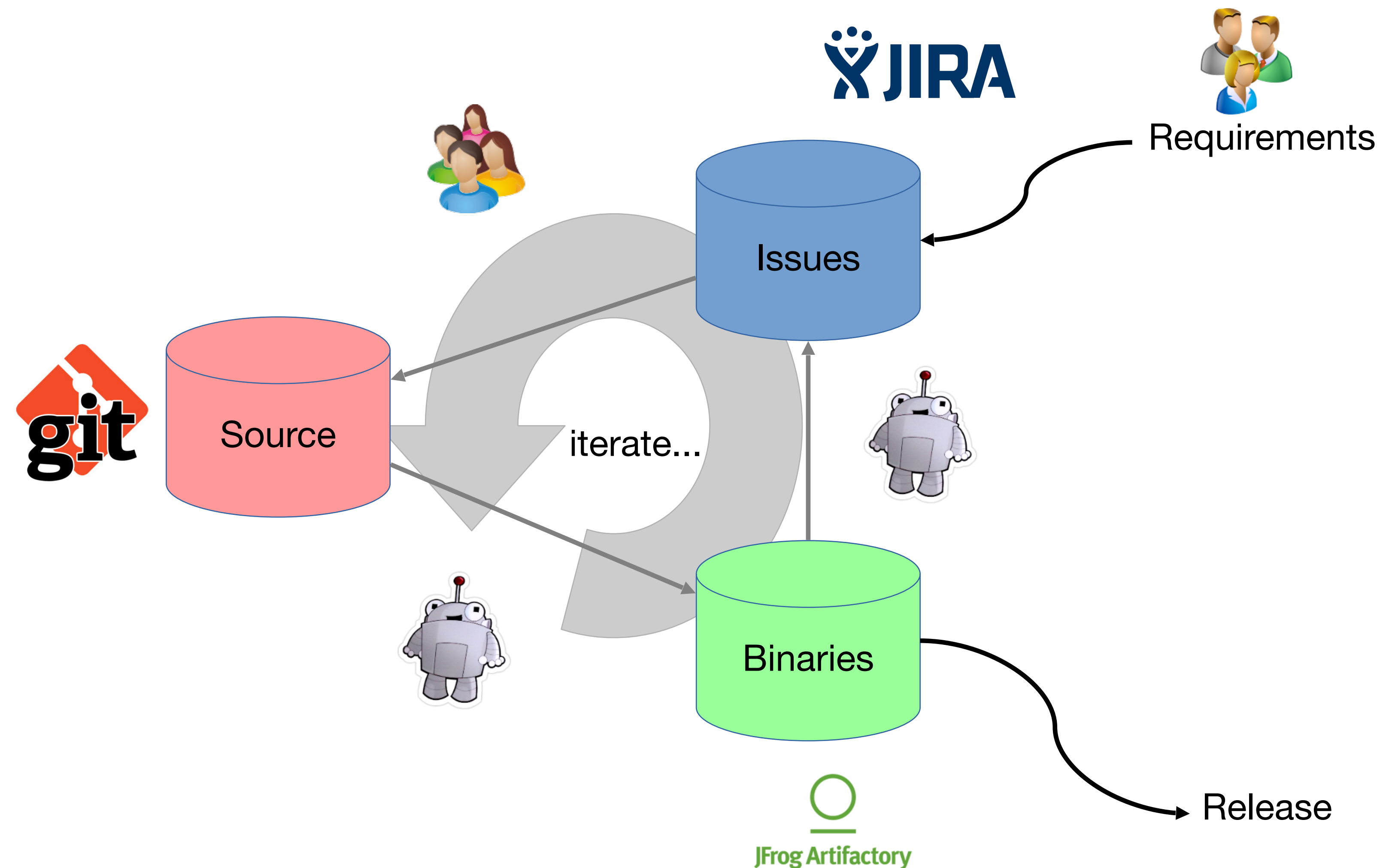


● Monitoring and Benchmarking

- Finding anomalies (performance, failures, ...) in complex applications and infrastructures

Yet another example...

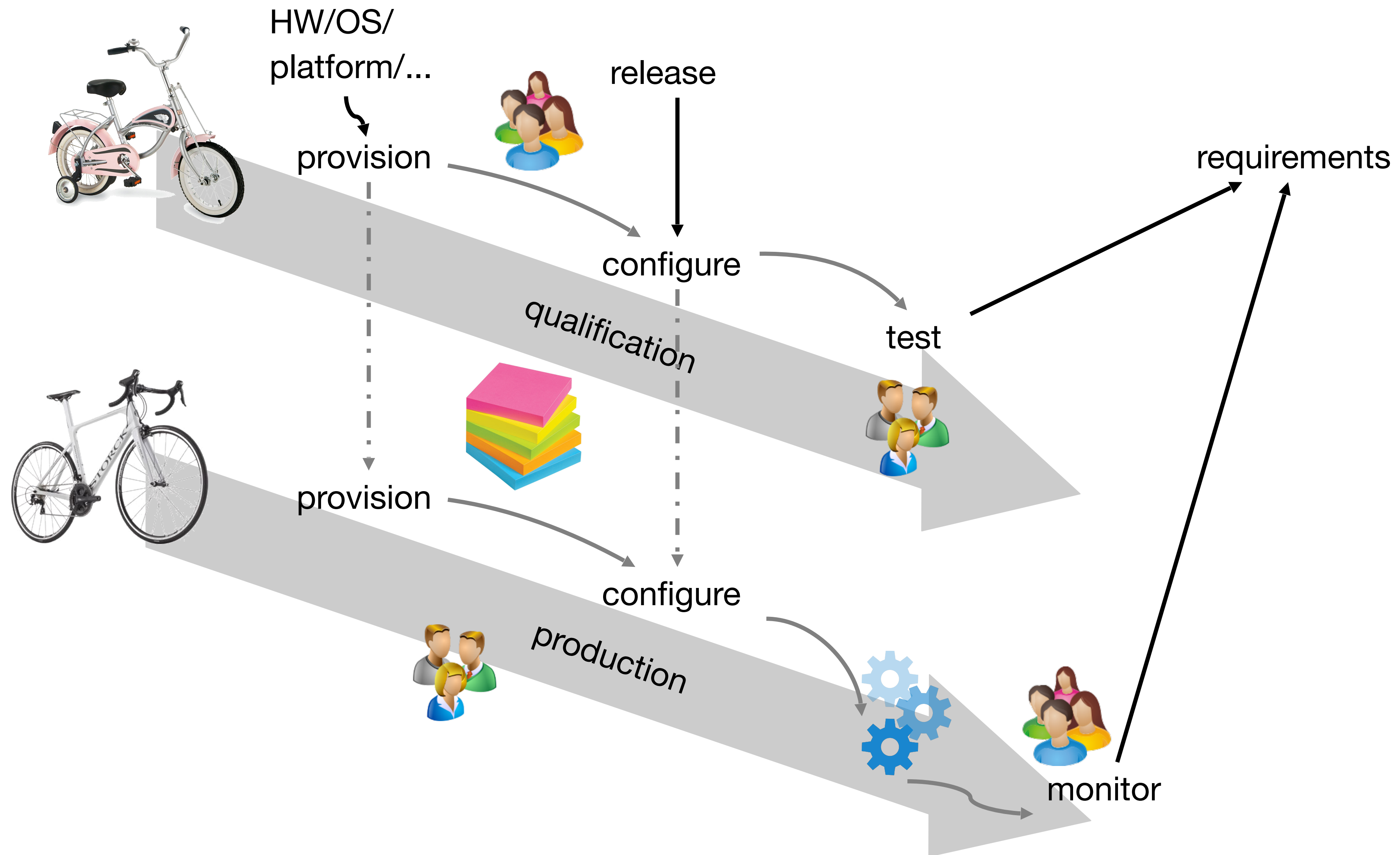
Agile development



Agile development

- ◎ Rests on formal (versioned) source, binary and documentation repositories
 - Unambiguous current state
 - Ability to back track
- ◎ Automated build and test
 - Fast feedback
- ◎ Fast iterations
- ◎ **Quick reaction to frequent small changes!**

Operations

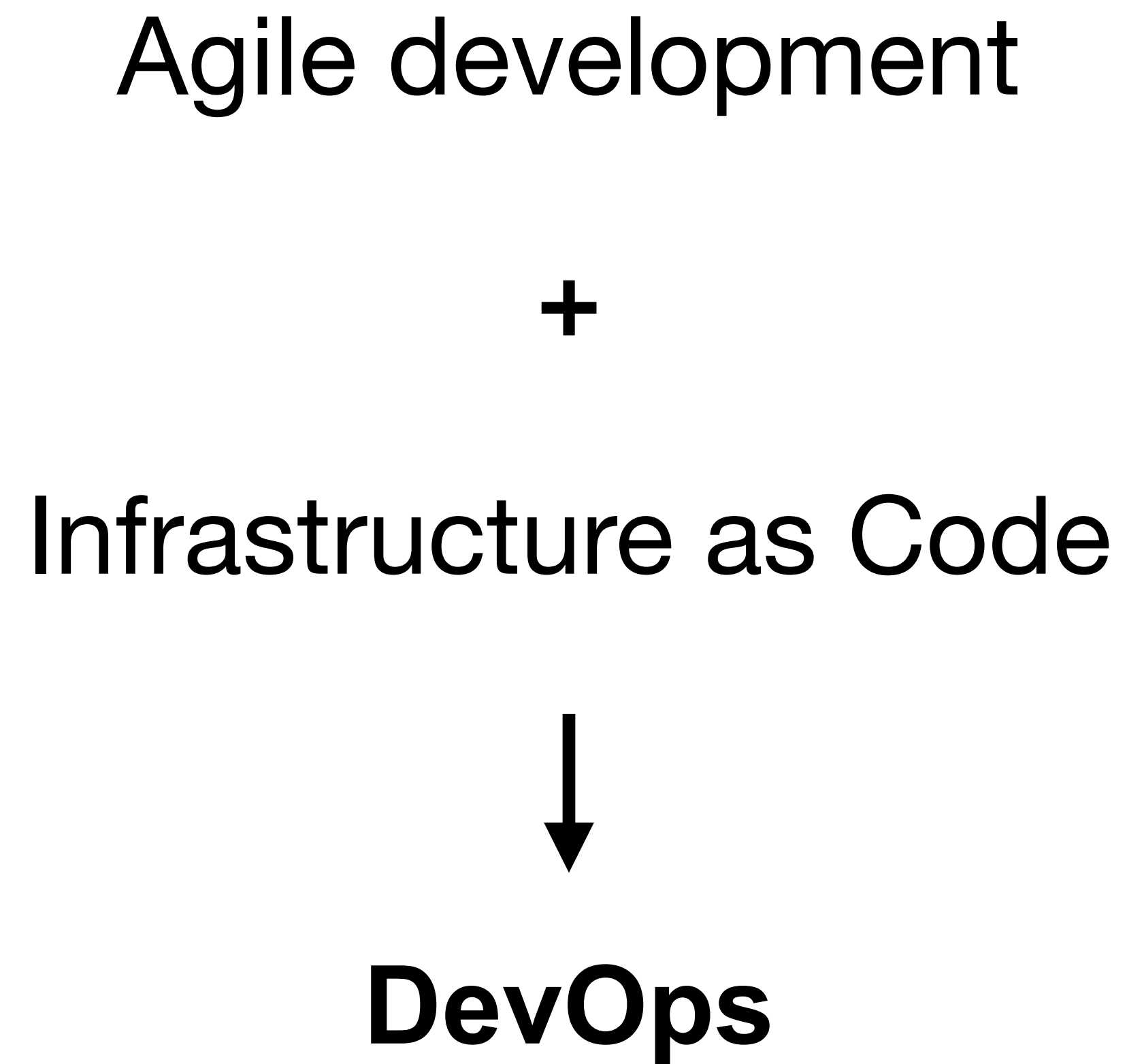


Operations

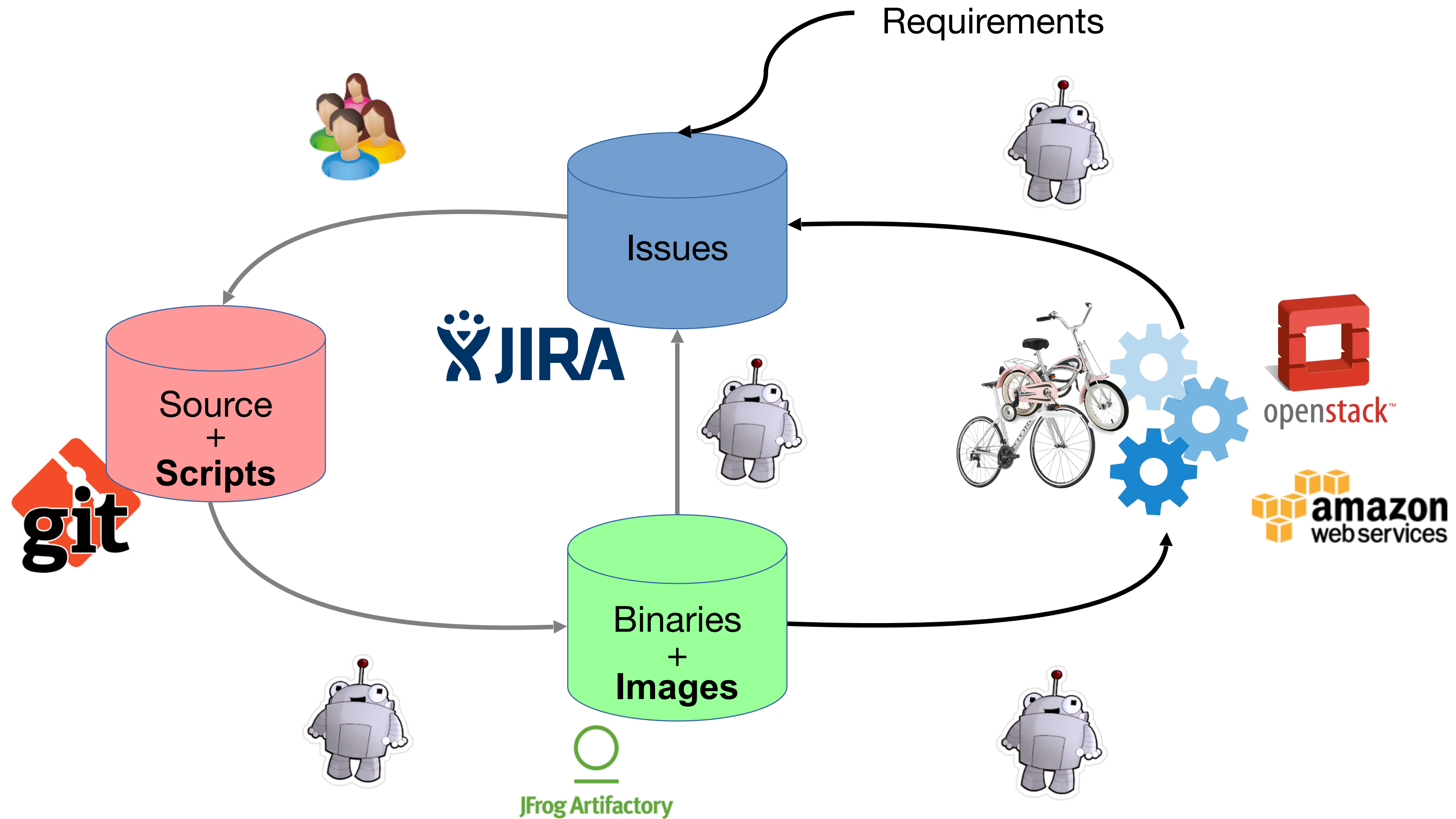
- Manual provisioning and configuration
- Informal communication between qualification and production stages
- Consequences:
 - Not repeatable or reproducible
 - Configuration drift leading to “Snowflake” servers (don't even look at it...)
 - Subjective monitoring

Infrastructure as Code

- Hardware provisioning with scripts
 - In contrast to: physically unboxing, plugging and configuring
- Software provisioning with scripts
 - In contrast to: clicking through setup wizards
- Configuration with scripts
 - In contrast to: clicking through control panels
- Include infrastructure scripts in agile process!



DevOps



DevOps

- Development and deployment are self-documenting and versioned
- Deployment is reproducible and repeatable
- Servers are disposable and consistent
- Supports fast, frequent, small changes!

Roadmap

● Distributed applications architectures & deployment

- Common distribution patterns
- Reliability and scalability
- Automatic provision and configuration

● Cloud computing

- Services and abstractions
- Infrastructure management
 - virtualization, storage, ...

● Monitoring and evaluation

- Monitoring frameworks and metrics
- Benchmarking

Assessment

- Project (50%) – minimum grade: 10 values
 - Checkpoint #1: 02/11/2025
 - Checkpoint #2: 30/11/2025
 - Final Report and Submission: 29/12/2025
 - Presentation: 12-17/01/2026
- Written exam (50%) – minimum grade: 8 values
 - 05/01/2026

Project

- Fill the following form by **September 28th**

- ▶ <https://forms.gle/GiFKSifntxsVtJnj7>
- ▶ We will use this information to add you to ASCN's GitHub organization
- ▶ Check your e-mail for an invitation after this date! The invite expires in 7 days.

- ASCN Github organization

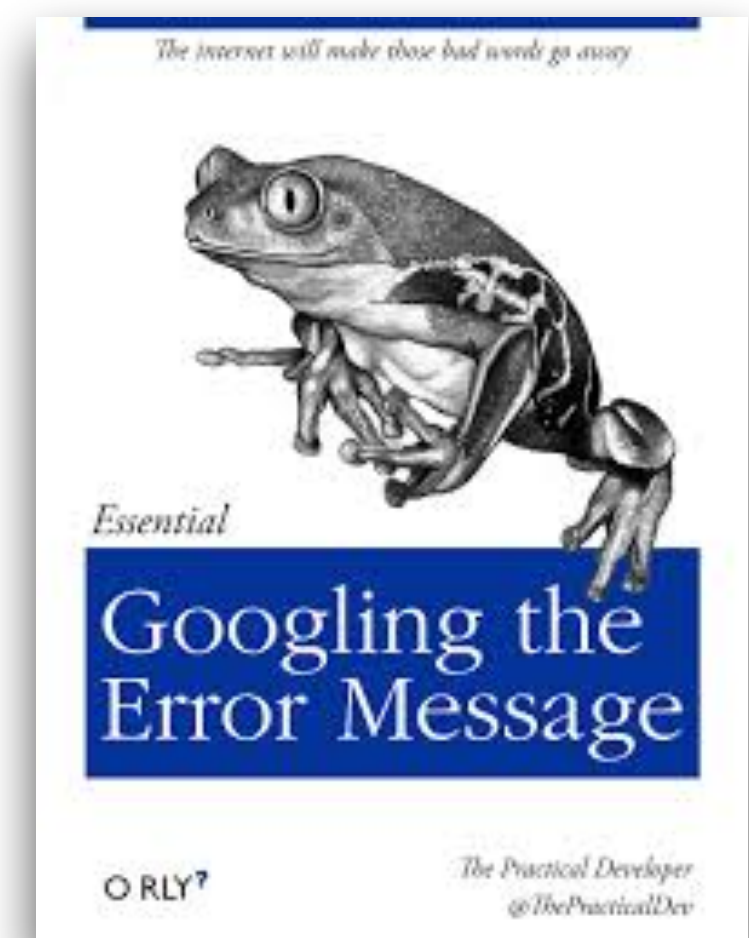
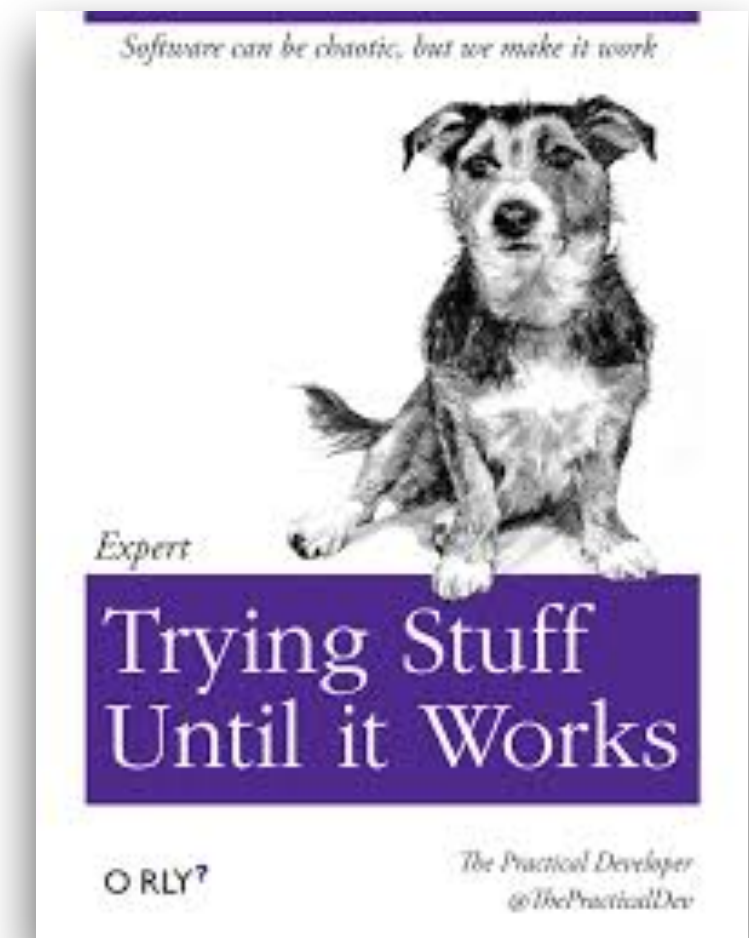
- ▶ When you are granted access, **carefully read** the “How to use the ASCN-UM organization” at the **Instructions discussion thread**

- Each group should have **5 elements**

- ▶ Start talking with your colleagues to find a group!
- ▶ The organization has a **Groups discussion thread** to find members/groups

Main References

- K. Morris. *Infrastructure as Code: Managing Servers in the Cloud*. O'Reilly, 2016
- M. Kleppmann. *Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems*. O'Reilly, 2017
- T. Erl, R. Puttini e Z. Mahmood. *Cloud Computing: Concepts, Technology and Architecture*. Prentice Hall, 2013
- R. Jain. *The art of computer systems performance analysis - techniques for experimental design, measurement, simulation, and modeling*. Wiley, 1991



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Questions?